U-Net

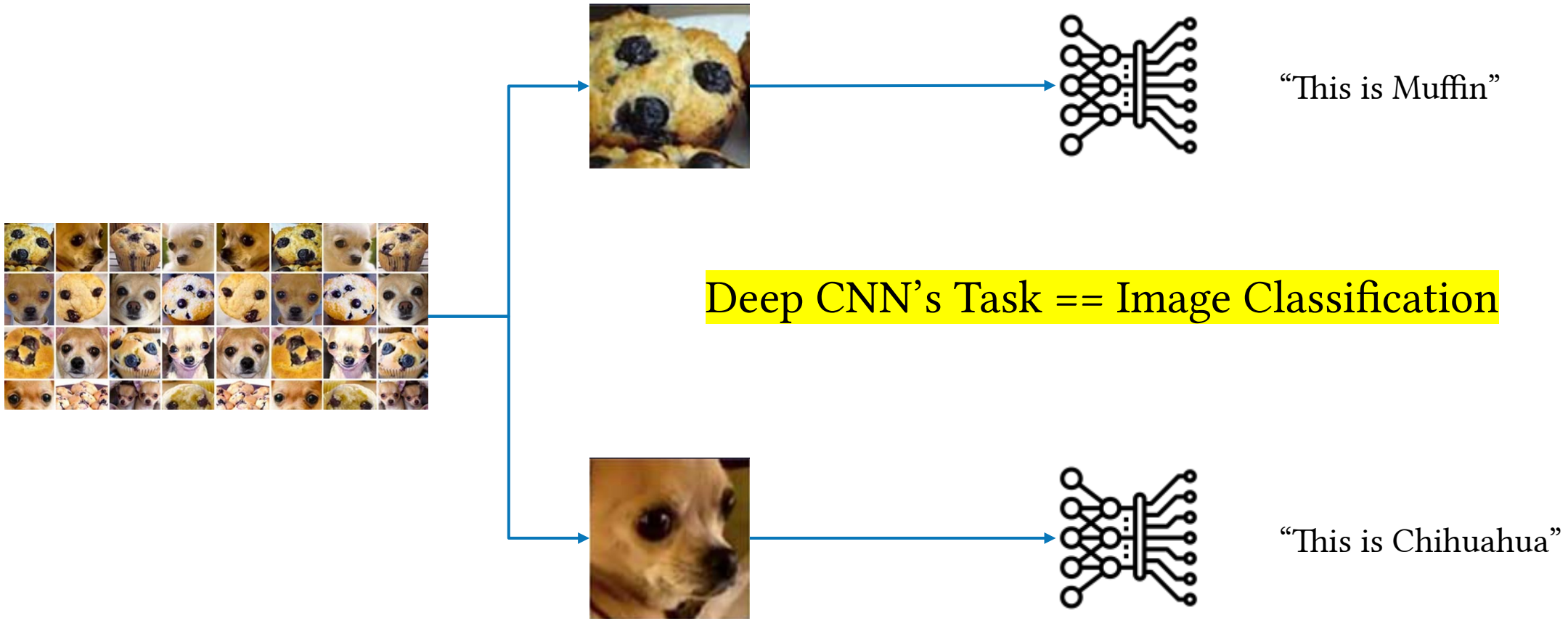
Convolutional Networks for Biomedical Image Segmentation

Ronneberger · Fischer · Brox | MICCAI 2015

2026.10.16 IRV-LAB

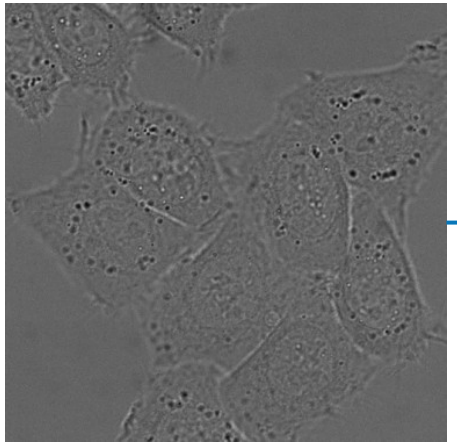
CNN's Task

(2026.10.15) **11** .



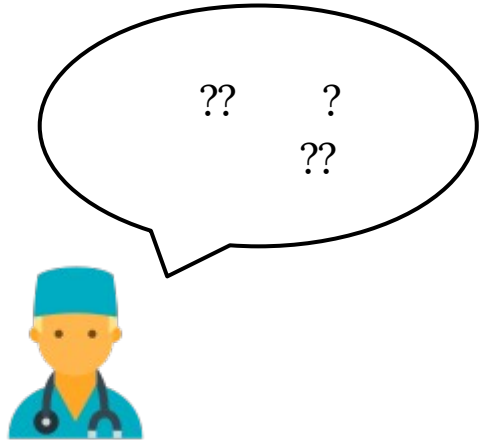
“ (Segmentation)”

, Biomedical Image Classification .



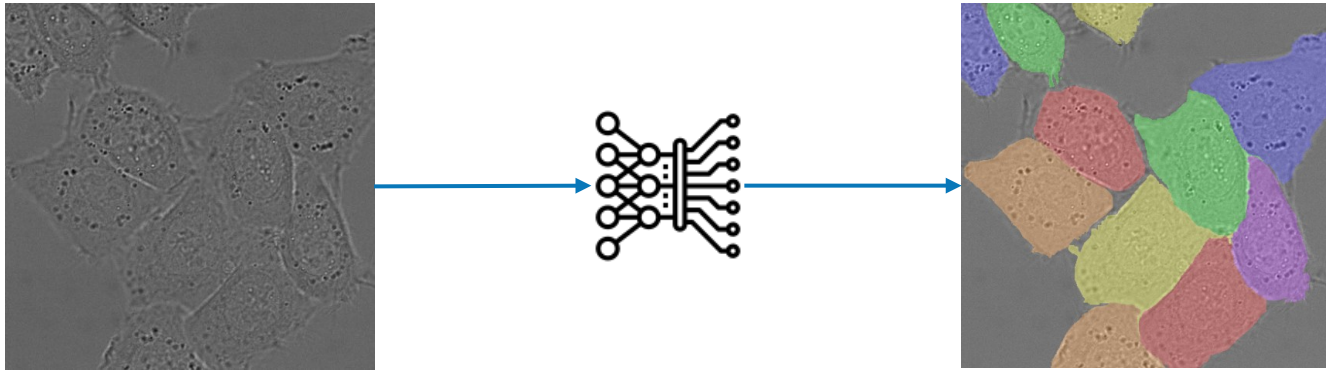


!!

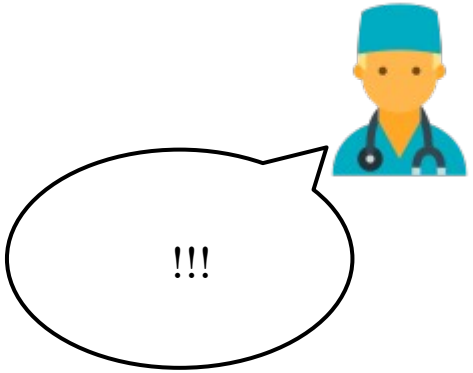


“ (Segmentation)”

, Biomedical Image .
, Classification .



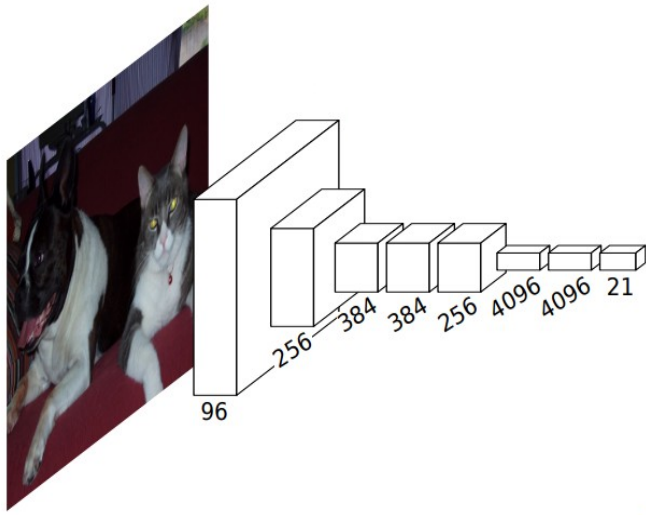
!!



- CNN: **class Label**
- Biomedical CNN: **Pixel** **class label**

Segmentation

CNN Segmentation ?



CNN segmentation .
, CNN feature map
segmentation .

Segmentation

, Segmentation .

sliding-window localize .

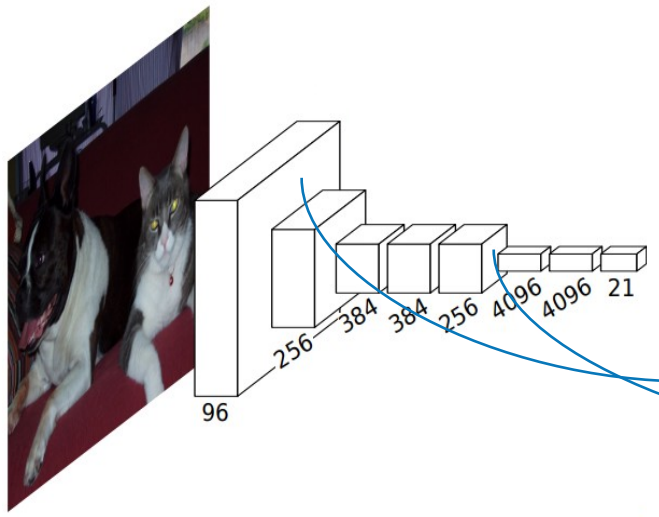
1. redundancy
2. localization accuracy use of context trade-off

patch size ↑: localization accuracy ↓
patch size ↓: network → “ .

Segmentation Core Idea

(Context)

(Localization)



- CNN : local high-resolution feature
- CNN : context-rich feature

FCN

U-Net 2014
"Fully Convolutional Networks for Semantic Segmentation" FCN .

FCN FCN .

U-Net FCN 2 .

1. Upsampling Convolution

   Conv feature map spatial resolution ,
   prediction .
   feature map segmentation Upsampling Convolution .

2. Skip Connection

   Upsampling downsampling ,
   Upsampling **CNN** ( ) .

FCN ① Upsampling Convolution

3×3 valid Conv

4×4 → 2×2

2×2 Up-Conv

$$I = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$$

$$K = \begin{bmatrix} k_1 & k_2 \\ k_3 & k_4 \end{bmatrix}$$

$$a_1 K = \begin{bmatrix} a_1 k_1 & a_1 k_2 \\ a_1 k_3 & a_1 k_4 \end{bmatrix}$$

4×4

$$\begin{bmatrix} a_1 k_1 & a_1 k_2 & a_2 k_1 & a_2 k_2 \\ a_1 k_3 & a_1 k_4 & a_2 k_3 & a_2 k_4 \\ a_3 k_1 & a_3 k_2 & a_4 k_1 & a_4 k_2 \\ a_3 k_3 & a_3 k_4 & a_4 k_3 & a_4 k_4 \end{bmatrix}$$

※ stride 2, kernel 2×2

FCN ② Skip Connection

upsampling **?**

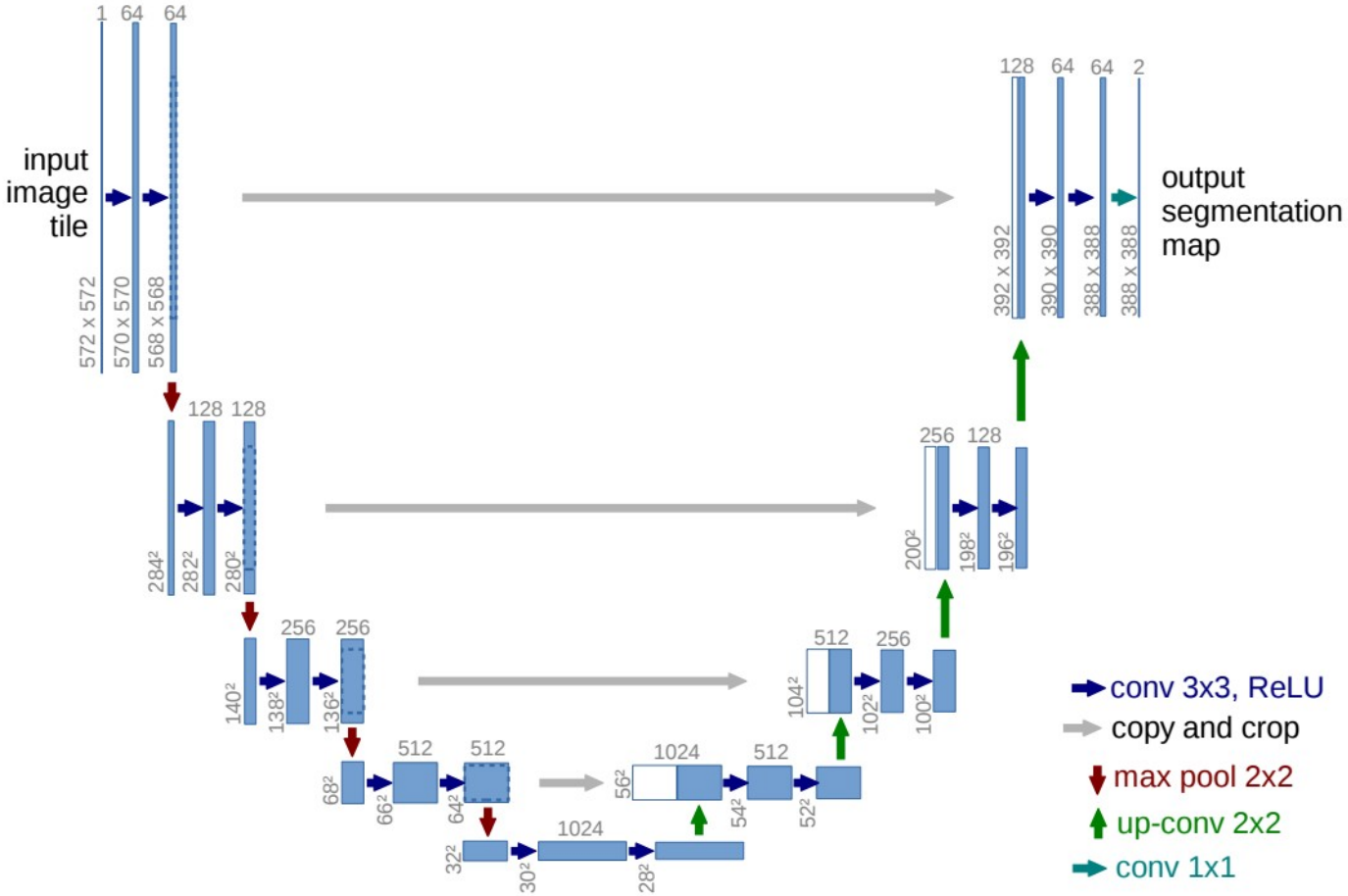


7 **1, 2, 4** **.**

Skip connection

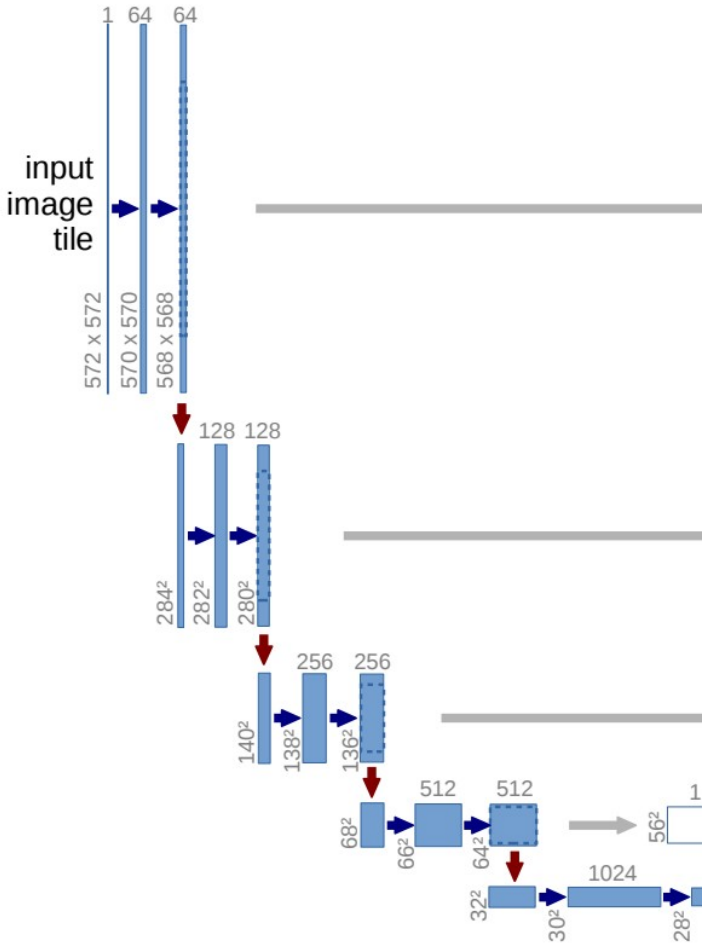
Encoder feature , upsampling .

U-Net



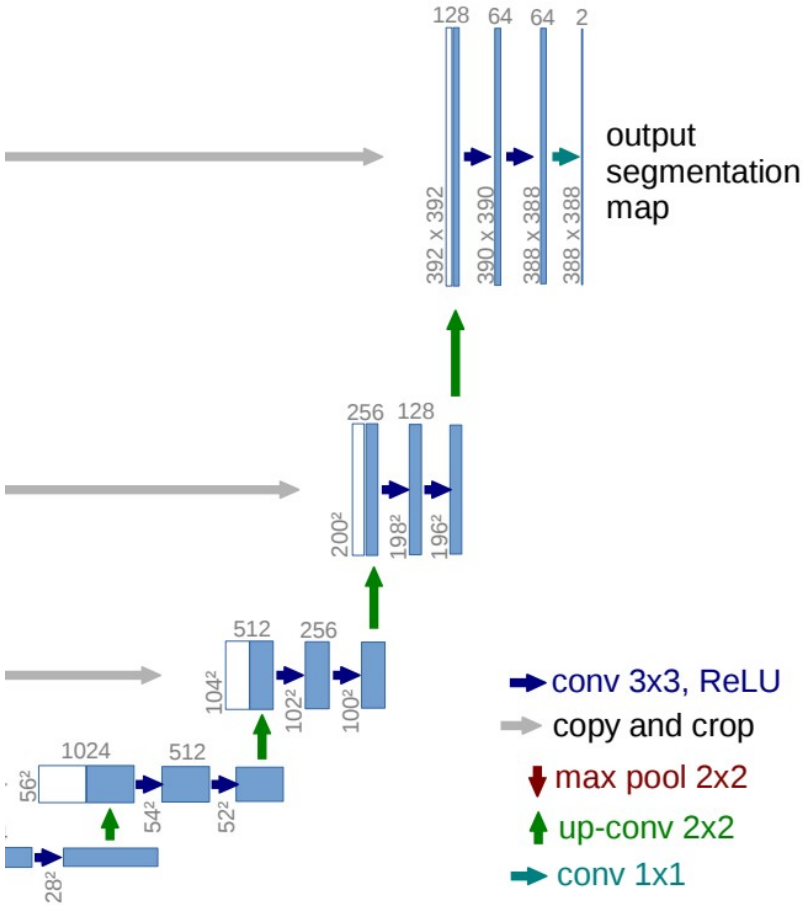
= skip connection (copy & crop)

Contracting path



- 3×3 valid convolution + ReLU
- 2×2 max pooling ½
- ×2

Expansive path



- 2×2 up-convolution: ↑, ½
- encoder feature crop concat
- 3×3 convolution + ReLU

1×1 convolution

K .

concat ?

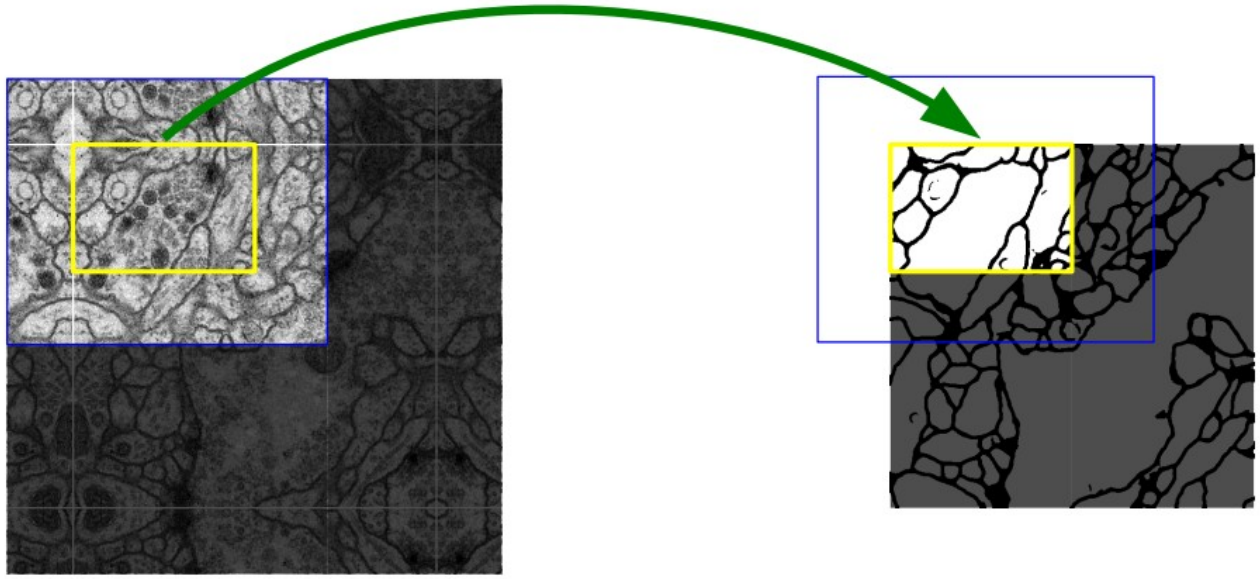
FCN skip element-wise addition .
, U-Net channel concat .

- Encoder: [, , texture]
- Decoder: [, ,]
  ,

- **addition**
  [+ , + , texture +]
  channel .
  channel .

- **concat**
  [, , texture, , ,]
  .
  channel .

: overlap-tile mirroring



: tile (overlap)

: (mirroring)

: tile, batch size 1, momentum 0.99

tile ?

valid conv input output .
tile (batch size 1) .

SGD · batch size 1

GPU tile .

Momentum 0.99

.

Pixel-wise softmax

x K .

$$p_k(x) = \frac{\exp(a_k(x))}{\sum_{j=1}^{K} \exp(a_j(x))}$$

- $a_k(x)$: x k logit
- $p_k(x)$: softmax
- $x \in \Omega$: segmentation map

Pixel-wise weighted cross entropy

class **loss** .

$$E = -\sum_{x \in \Omega} w(x) \log p_{l(x)}(x)$$

w(x)

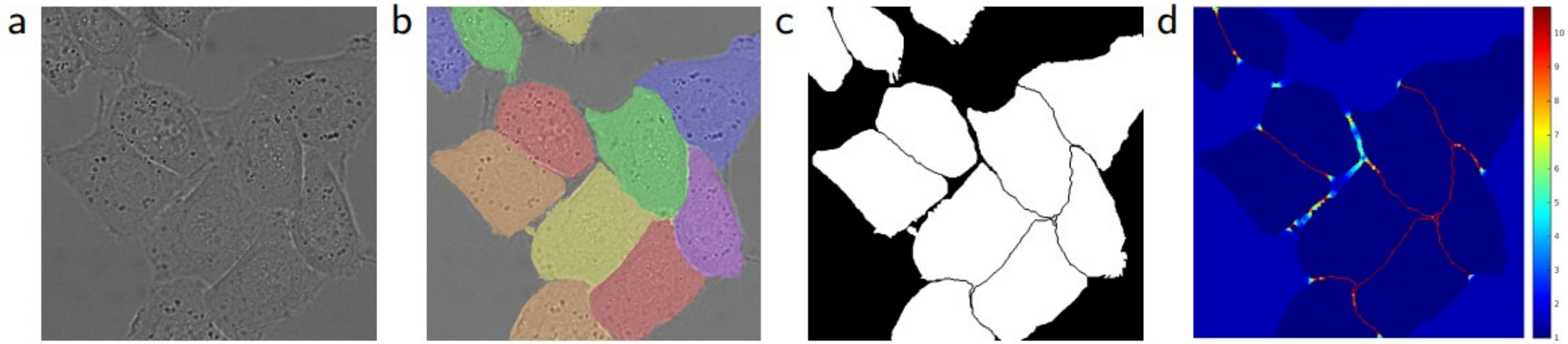
$$\mathrm{CE} = -\sum_{k=1}^{K} y_k \log(p_k)$$

$p_{l(x)}(x)$ x label

:

.

weight



a b c d weight map

$$w(x) = w_c(x) + w_0 \exp\left(-\frac{(d_1(x) + d_2(x))^2}{2\sigma^2}\right)$$

d_1, d_2↑ | $w_0 = 10$, $\sigma \approx 5$ px

annotation Data Augmentation

.

Elastic deformation

- 3×3 coarse grid Gaussian
- 10 px, bicubic interpolation
- shift · rotation · gray-value

: ( ) · contracting path dropout

biomedical segmentation task

Dataset			
EM challenge		30 ·	Warping / Rand / Pixel error ↓
PhC-U373		35 ·	IoU ↑
DIC-HeLa	DIC	20 ·	IoU ↑

task U-Net .

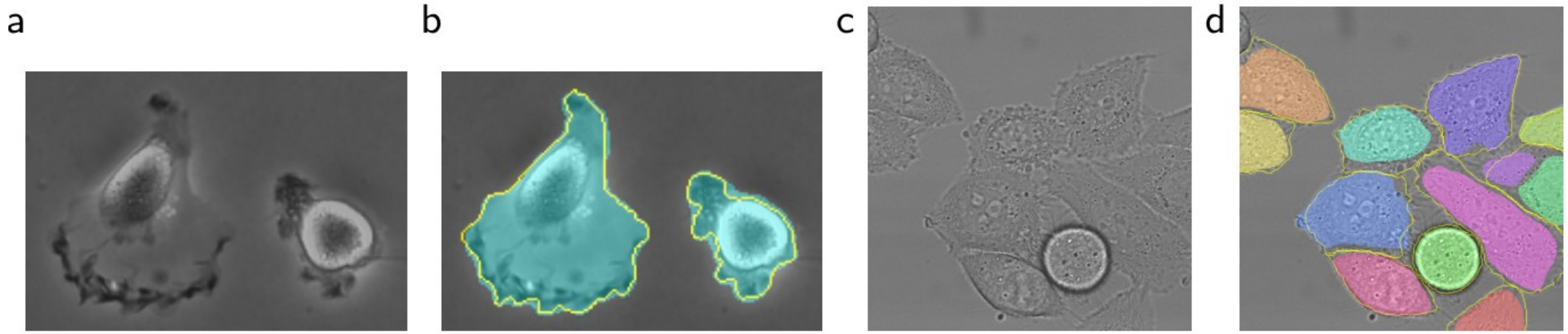
EM challenge

Method	Warping ↓	Rand ↓	Pixel ↓
Human values	0.000005	0.0021	0.0010
U-Net	**0.000353**	**0.0382**	**0.0611**
DIVE-SCI	0.000355	0.0305	0.0584
IDSIA	0.000420	0.0504	0.0613

Warping 1 . Rand·Pixel DIVE-SCI .

U-Net 7 forward pass .

Cell tracking:



a·b PhC-U373 ( ) c·d DIC-HeLa ( )

: | :

Cell tracking challenge

Method	PhC-U373 IoU ↑	DIC-HeLa IoU ↑
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	—
Second-best 2015	0.83	0.46
U-Net (2015)	**0.9203**	**0.7756**

U-Net **2015** **IoU** **.**

U-Net

expansive path · **feature concat**

elastic deformation

loss weight

overlap-tile **mirroring**

, . .

- ablation
- 2D biomedical task . MRI, CT 3D
- Momentum 0.99 batch sampling ,
- valid conv crop·tiling
- 2015

Conclusion

U-shaped

, decoder

▼

!!(Diffusion Policy IMLE Policy 1D U-Net)

Elastic deformation loss

1D U-Net



Conv1D (kernel=3) t2–t4: → →

feature ( ) t0–t5: → →

T=16 → 8 → 4 → 8 → 16

Skip concat: encoder feature decoder

t3 gripper timestep

·T=16 . local + context skip .

Diffusion·IMLE 1D U-Net

Diffusion Policy

$$\hat{\epsilon} = \epsilon_\theta(O_t, A_t^k, k)$$

A^k_t: k noisy action

O_t: (·) feature

k: denoising

O_t·k embedding FiLM

U-Net → timestep noise

denoising

IMLE Policy

$$\hat{A}_t = T_\theta(z, \mathcal{O}_t)$$

z: Gaussian latent

$\mathcal{O}_t$: feature

diffusion k

feature FiLM

U-Net → action

z

: . U-Net 1 .